

Name:

Module 2 Assignment: Law Enforcement Equipment

/58

Lesson A: The Bullet-Resistant Vest

Use the following information to answer the next question.

Various Molecular Traits

1. Natural polymer
2. Synthetic polymer
3. Radial composition
4. Square composition
5. Contains hydrogen bonds
6. Contains only single bonds
7. Fibres oriented in a circular pattern
8. Fibres oriented in a parallel arrangement

1. Four molecular traits of Kevlar, as shown above, include _____ , _____ , _____ , _____ , and _____. (2 marks)

2, 3, 5, 8

Use the following information to answer the next question.

Various Descriptions Related to Bullet-Resistant Vests

1. Invented unintentionally by a chemist in the US
2. Durable and lightweight; ideal for police officers
3. Can absorb energy delivered by rifle rounds or knives

2. Match each numbered description with the correct material or object as given below: (3 marks)

Material/Object	Description
Soft body armour	2
Kevlar	1
Hard body armour	3

Name:

3. Identify two advantages and two disadvantages of Kevlar in bullet-resistant vests. (4 marks)

Any two of the following advantages are acceptable for 2 marks

- diffuses the force of low and medium velocity bullet impact
- absorbs the kinetic energy of low and medium velocity bullets
- 5 times stronger than steel
- lightweight
- flexible
- does not rust or corrode
- absorbs vibrations

Any two of the following disadvantages are acceptable for 2 marks:

- expensive
- breaks down in sunlight
- does not stop high-velocity rifle sounds
- does not prevent injury from bladed weapons
- breaks down when washed

4. Explain what a trauma plate is and why it is inserted into a bullet-resistant vest. (2 marks)

A trauma plate is a ceramic or metal plate surrounded by several layers of kevlar. 1 mark
It helps prevent injury to the heart, lungs, and ribs that may arise from "blunt force trauma", a type of injury sustained by the sudden force of a bullet pushing but not quite completely penetrating a vest. Blunt force trauma can cause severe bruising, cracked ribs, or even death. 1 mark

5. Identify two factors that can cause the breakdown of Kevlar over time, and describe how bullet-resistant vests have been designed to prevent this (3 marks)

Any two of the following are acceptable for 2 marks:

- ultraviolet rays/sunlight exposure
- dry-cleaning agents
- bleach
- repeated washing
- moisture
- bullet-resistant vests are built with layers of Kevlar protected by a fabric overting to prevent exposure, damage, and the need to clean the underlying Kevlar

6. Why was the testing of the first bullet-resistant vest designed for police officers controversial? (2 marks)

The inventor of the bullet-resistant vest, Richard Davis, risked his life when he shot himself while wearing the vest to demonstrate its efficacy.

Name:

Lesson B: The Conducted Energy Device (CED)

Use the following information to answer the next question.

Steps Occurring inside a Conducted Energy Device (CED):

- 1. The trigger of the CED is activated.**
- 2. Electric current flows into a step-up transformer.**
- 3. The voltage of the power supply increases.**
- 4. The oscillator converts energy from the batteries into a stable electrical resonant circuit.**

7. The correct sequence of the steps shown above occurring in a functioning CED are _____, _____, and _____. (2 marks)

1, 4, 2, 3

Use the following information to answer the next question.

Various Descriptions of Electroshock Devices:

- 1. Wireless projectile that can be fired from a 12-gauge shotgun**
- 2. Handled electroshock weapon used by police to subdue subjects**
- 3. Stand-alone device that observes, warns, incapacitates, and retains intruders**

8. Match each electroshock device description numbered above with the appropriate device in the chart below: (3 marks)

Device	Description
CED	2
XREP	1
TRAD	3

9. Describe how the symptoms of being contacted with the CED for one-second differ from the symptoms of a five-second CED contact. (2 marks)

Name:

A one-second contact with a CED will startle the attacker and cause a certain amount of pain associated with nerve and muscle disruption. (1 mark)
An electrical charge lasting in cycles of five seconds causes a continuous contraction of muscles throughout the subject's body, resulting in a loss of balance and muscle control as well as temporary cognitive disorientation. (1 mark)

10. Identify three advantages to the use of CEDs by police officers. (3 marks)

Any three of the following disadvantages are acceptable for 3 marks

- has lowered number of officer-involved firearms-related shootings
- can be used as a contact weapon or as a projectile weapon
- immobilizes suspect without causing long-term injuries
- can be deployed quickly
- does not break bones
- does not cause soft tissue damage
- the sight of it or warning of it subdues many suspects without having to use it

11. Identify three factors that affect the electrical current experienced by a subject hit by a CED (3 marks)

Any 3 of the following are acceptable for 3 marks.

- target's resistance
- moisture
- body salinity
- clothing
- CED internal circuitry
- battery conditions

12. Describe two situations that warrant the use of a CED by a police officer. (2 marks)

Any two of the following are acceptable for 2 marks:

- a subject is resisting arrest
- the subject is brandishing a weapon
- the subject has a history of violent behaviour

Lesson C. Pepper Spray and Tear Gas

Use the following list to answer the next questions.

Name:

Various Traits of OC Pepper Spray and CS Tear Gas

1. **Natural compound**
2. **Synthetic compound**
3. **Irritates the stomach**
4. **Irritates the eyes and nose**
5. **Soluble in fats and oils**
6. **Soluble in an alkaline solution**
7. **Used in riot control**
8. **Used upon unruly individuals**

13. a. Four unique traits of OC in pepper spray include ____, ____, ____ and _____. (2 marks)

1, 4, 5, 8

b. Four unique traits of CS in tear gas include ____, ____, ____, and _____. (2 marks)

2, 4, 6, 7

Note: The same trait can be used in both part a and b.

14. Why is a pepper spray with a lower concentration of OC often more effective than a pepper spray with a higher OC concentration? (2 marks)

- Pepper sprays with low OC concentrations are less viscous or lighter than sprays containing higher concentrations of OC. (1 mark)
- The lighter the liquid in a spray, the faster it will penetrate the mucous membranes (1 mark)

15. Explain why tear gas is often dispersed using grenades. (2 marks)

Tear gas grenades are useful in propelling tear gas over a great distance into large crowds. (2 marks)

16. Explain why OC pepper spray and CS tear gas are both considered “less-than-lethal” weapons (2 marks)

Both pepper spray and CS gas are considered to be “less-than-lethal” weapons because they have only temporary effects upon individuals and individuals can be subdued without using a gun.

17. Identify one similarity and one difference between pepper spray and tear gas (2 marks)

Any one of the following similarities is acceptable for 1 mark

Name:

- Both are considered to be less-than-lethal.
- Both are released through hand-held canisters
- Both cause only short-term physiological effects.
- Capsaicin and CS are solids.
- Both irritate the eyes, nose, mucous membranes, and skin.

Any one of the following differences is acceptable for 1 mark

- Tear gas causes severe coughing/irregular breathing.
- Tear gas washes out with alkaline solution (such as baking soda solution).
- Pepper spray washes out with fat and/or oil solution (such as milk or dish detergent).
- Tear gas effects last less than 15 min; pepper spray effects last 30-45 min.
- Tear gas is used to subdue large crowds; pepper spray is used to subdue unruly individuals.
- OC never dissolves in water; CS gas eventually dissolves in water.
- Pepper spray is from a natural compound but tear gas is synthetic.

Lesson D. Case Study Involving Law Enforcement Safety Equipment

Use the following list to answer the next question:

Possible Descriptions of the North Hollywood Shootout

- 1. A bank security guard called police**
- 2. A witness outside the bank contacted the police**
- 3. Both suspects were shot and killed**
- 4. Both suspects survived and were apprehended**
- 5. The case involved 32 police officers**
- 6. The case involved 370 police officers**
- 7. Each suspect was wearing soft body armour**
- 8. Only one of the suspects was wearing body armour**

18. The four descriptions numbered above that describe the North Hollywood Shootout are ____, ____, ____, and _____. (2 marks)

2, 3, 6, 7

19. In the North Hollywood Shootout, explain why police did not use pepper spray, tear gas, or police canines to subdue the two suspects in this case (3 marks)

Tear gas or pepper spray was not used because both suspects were moving continually outdoors and firing automatic weapons. (2 marks)
Any police dog that may have attempted to capture the suspects would have likely been

Name:

shot immediately by the suspect with high-powered weapons. (1 mark)

Use the following list to answer the next question:

Possible Events that occurred in the Party Out of Control Incident

1. The hall is set on fire
2. A fist fight begins
3. The hall owner calls the police
4. The crowd develops a mob mentality
5. Several hall patrons make threatening gestures and insults
6. Several officers discharge pepper spray into the crowd
7. A police supervisor uses a fogger to disperse tear gas into the crowd
8. A police supervisor uses a fogger to disperse pepper spray in the crowd

20. The correct sequence of events that occurred in the Party Out of Control Incident is ____, ____, ____, and _____. (2 marks)

5, 2, 4, 8

Use the following list to answer the next question:

Possible Events that occurred in the Carl Rowland Incident.

1. Carl Rowland fled from police to Florida
2. Carl Rowland ran from authorities in Georgia
3. Carl Rowland's wife was sexually assaulted
4. Carl Rowland's girlfriend was strangled to death
5. Rowland was pepper sprayed and taken into custody
6. Rowland is tasered and lowered from the crane
7. Rowland pulled a knife on a group of construction workers
8. Rowland took a crane operator hostage at a construction site

21. The correct sequence of events numbered above that occurred in the Carl Rowland Incident ____, ____, ____, and _____. (2 marks)

4, 2, 7, 6

Name:

22. In the Carl Rowland Incident, why did the police lure the suspect before apprehending him (3 marks)

Police needed to lure the suspect from the top of the crane because they needed to get close enough to taser and apprehend him.

23. Provide 3 reasons why the handling of the Party Out of Control Incident was an effective use of non-lethal force. (3 marks)

Any 3 of the following are acceptable for 3 marks
-no long-term injuries to partygoers
-fast and effective way to disperse crowd
-did not cause any damage to scene
-prevented further damage to the hall building